Hari Krishna T

AI Engineer

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Experience

AI Engineer Intern, *Tech Aegis AI* 06/2026 – Present
Hyderabad, India

- Build and ship **AI automation systems daily** — LinkedIn outreach, Sales CRM, and Google Blogs / Articles & Ads content automation — automating 8+ business workflows via LLM-driven pipelines and n8n orchestration.
- Experienced with **voice AI agents** (STT → LLM → TTS) — implementing **response streaming**, **prefix caching**, and reducing **LLM tool calls** to lower conversation latency.
- Leverage **Claude Code** as a force multiplier — authoring custom **subagents, skills, hooks, and MCP servers** — cutting feature development & **delivery time by ~40%.**
- Design, build, and **evaluate production RAG systems** (hybrid semantic + BM25 retrieval, re-ranking), improving retrieval quality measured via Ragas (faithfulness, answer relevancy, context precision).
- Run **LLMOps & observability** — tracing, token/cost tracking, and automated evaluation gates — using Langfuse and LangSmith.
- **Experienced with Implementing Agent Security** & Deploy services to Vercel and AWS (EC2 / Bedrock), containerized with Docker for repeatable, reliable releases.

Skills

AI FRAMEWORKS
- LangChain: Chains, Memory, Agents
- LangGraph: Multi-Agent Systems, Workflow Design
- Hybrid RAG: Retrieval-Augmented Generation, Document Retrieval, Semantic Search & BM25 Keyword Search
- Vector DB: Qdrant, ChromaDB, FAISS

LLMOps & Observability
Langfuse (tracing, cost, eval), LangSmith, Ragas evaluation, prompt registry & self-improvement loops, Guardrails AI

Tools & Deployment
Docker, n8n, Git, GitHub, AWS EC2, Render, Hugging Face (Model Hosting), Streamlit Cloud,Neon, Kaggle GPU

AI Dev Productivity
Claude Code (Subagents, Skills, Hooks, MCP servers) — building agentic dev workflows to cut delivery time

Python
Data Structures , Control statements , OOP, Functions

GENERATIVE AI
Large Language Models (LLMs), Prompt Engineering, OpenAI GPT, Google Gemini, AWS Bedrock, Groq API, Hugging Face Transformers (BERT, GPT, T5), Tokenization, Pipelines, MCP , PyTorch , QLoRA / LoRA Fine-Tuning, PEFT, Model Evaluation ,RLHF, SFT, A2A Protocols

SQL
Database & Table Creation, CRUD Operations, SQL Joins, Data Retrieval, ALTER, DROP, TRUNCATE

ML & DL Foundations
Machine Learning (supervised/unsupervised), Neural Networks, Transformers, NLP (embeddings, tokenization, sentiment analysis)

FastAPI & Backend
FastAPI: REST API Development, Routing Request Handling, Pydantic Models, API Integration, Backend Development, JSON Handling

Projects

Atlas — Autonomous Due-Diligence Desk, *Tech Stack: Python, LangGraph, LangChain, Groq (GPT-OSS-120B),FastAPI, Qdrant, BM25, bge Embeddings, Ragas, Langfuse, Next.js, Docker, MCP*
- Architected a production-grade multi-agent due-diligence system using LangGraph StateGraph with a Supervisor → Specialist workflow — four specialists (Fundamentals, News/Sentiment, Risk, Market) researching in parallel over shared SQLite-checkpointed state, followed by a Bull vs. Bear debate and a Synthesizer/Judge producing an investor-grade cited report with a confidence score.
- Developed an **agentic Hybrid RAG pipeline** — BM25 + bge dense embeddings via EnsembleRetriever (RRF), a CrossEncoder re-ranker, and a **CRAG self-correction loop** over Qdrant / InMemory, with an **MCP tool server** (web_search, SEC EDGAR, stock_data, company_news) for live evidence.

- **Engineered four memory layers** — Working (graph state), Semantic (RAG), Episodic (SQLite + vectors of past runs), and Procedural (playbooks) — with a Summarizer agent (cheaper GPT-OSS-20B) distilling past runs into durable facts.
- **Built a self-improving LLMOps loop:** automatic **Ragas evaluation** (Faithfulness, Answer Relevancy, Context Precision) → quality gate → prompt rewrite → re-eval → release via a versioned prompt registry.
- **Integrated Langfuse for full-trace observability**, token/cost tracking, and Ragas dashboards; enforced a grounding guardrail so no report claim exists without a retrieved-evidence citation.
- Built a FastAPI + SSE streaming backend with a live Next.js "war-room" frontend; runs zero-cost on Groq's free tier with local bge embeddings, containerized with Docker — frontend deployed on Vercel, backend on Render.

MediaMind — Autonomous Media Intelligence Platform ,
Tech Stack: Python, LangGraph, LangChain, Groq (GPT-OSS-120B), ChromaDB, BM25, Streamlit, Render, MCP

- Architected a production-grade multi-agent AI platform using LangGraph StateGraph with a Supervisor → Specialist workflow for intelligent task routing across Summarization, Highlight Generation, and Social Media content agents.
- Developed a **Hybrid RAG pipeline** combining ChromaDB semantic retrieval (60%) and BM25 keyword retrieval (40%) with ranking, merging, and deduplication; integrated autonomous MCP-style tool-calling (Wikipedia, Web, DuckDuckGo, YouTube Transcript).
- Built a real-time YouTube transcript ingestion pipeline (youtube-transcript-api) with automatic extraction, embedding generation, and dynamic ChromaDB re-indexing, plus a memory-enabled multi-session chat system in Streamlit.
- **Integrated Groq GPT-OSS-120B with task-specific temperature** optimization (0.0 routing, 0.3 summarization, 0.75 creative) and RAGAS evaluation (Faithfulness, Answer Relevancy, Context Precision, Hallucination).
- **Integrated Guardrails AI for safety and Langfuse for tracing/analytics;** containerized with Docker, deployed on AWS EC2, optimized for zero-cost local embeddings (all-MiniLM-L6-v2).

Enterprise Text-to-SQL — Fine-Tuned LLM with Self-Correction , *Tech Stack: Python, PostgreSQL, Qwen3-8B, QLoRA/PEFT, Transformers, FastAPI, Docker, SQLGlot, Neon, Render, Hugging Face*

- **Fine-tuned Qwen3-8B using QLoRA** on a free Kaggle T4 GPU — training only **43.6M parameters (0.917%)** of an 8B model — raising SQL execution accuracy from **43.71% to 70.86%** on 453 unseen questions.
- **Built the benchmark myself**: generated a 12-table PostgreSQL database (328K rows) and **3,087 question–SQL pairs**, split by question pattern so no pattern appears in both training and testing (**zero overlap**).
- **Scored by executing SQL, not comparing text** — each query runs against the real database and result sets are compared, so differently-written but correct queries still count as correct.
- **Improved hardest queries most**: multi-join accuracy rose **7.6% to 54.9%**; classified all 132 remaining failures by root cause to separate logic errors from formatting differences.
- **Ran a 5-configuration ablation study** and reported a negative result: schema retrieval *reduced* accuracy by 9.3 points, so it was removed.
- **Added a self-correction loop** that feeds the database error back to the model — executable SQL rose **94.5% to 98.2%**, hallucinated tables/columns fell **3.75% to 0.66%**.
- **Shipped a FastAPI service** with 3 security layers (read-only DB user, query timeouts, SQL validation) — **166 automated tests**, deployed on Render + Neon.
- **Published model, dataset and live GPU demo to Hugging Face.**

Education

Gen AI, *Innomatics Research Labs* 01/2026 – 06/2026
Coursework: LLMs, LangChain, LangGraph, Hybrid RAG, RAG Evaluation, Agentic AI, MCP, Langfuse, FastAPI, Docker. Hyderabad, India

Bachelor of Technology (B.Tech) - Artificial Intelligence & Data Science, 10/2022 – 04/2026
Aditya College of Engineering Madanapalle, India
Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Python, DSA, SQL.

Certificates

Certification in Generative AI
Innomatics Research Labs

Python Developer
Innomatics Research Labs

Artificial Intelligence
EI Systems

Introduction to Data Science
Infosys | Springboard

Certification in AI Data Quality Analyst
Rooman Technologies Pvt Ltd, Bangalore